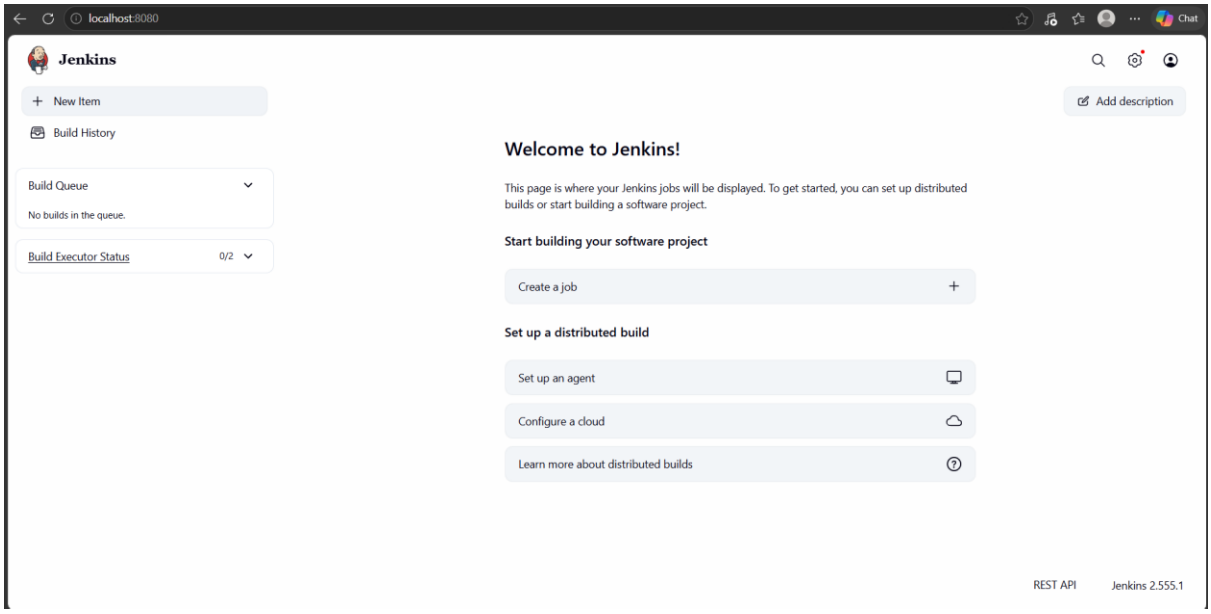
Name: Kashish Paryani Batch: T21 Roll No: 83

EXPERIMENT 5

Aim: To Build the pipeline of jobs using Maven / Gradle / Ant in Jenkins, create a pipeline script to Test and deploy an application over the tomcat server

Theory: Continuous Integration and Continuous Deployment (CI/CD) is a software development practice that automates the process of building, testing, and deploying applications. Jenkins is a widely used open-source automation server that enables developers to create pipelines for integrating code changes, running tests, and deploying applications efficiently. In this experiment, build tools such as Apache Maven, Gradle, or Apache Ant are used to compile source code, manage dependencies, and package the application. A Jenkins pipeline is defined using a script (commonly called a Jenkins file), which specifies stages such as Build, Test, and Deploy. During the build stage, the selected tool (Maven/Gradle/Ant) compiles the project and generates an artifact (e.g., WAR file for web applications). In the testing stage, automated test cases are executed to ensure code quality. Finally, in the deployment stage, the generated artifact is deployed to a web server such as Apache Tomcat, which hosts and runs Java-based web applications.

This pipeline-based approach ensures faster development cycles, early detection of errors, and consistent deployment practices. By integrating build tools with Jenkins and deploying on Tomcat, organizations achieve automation, scalability, and reliability in their software delivery process.



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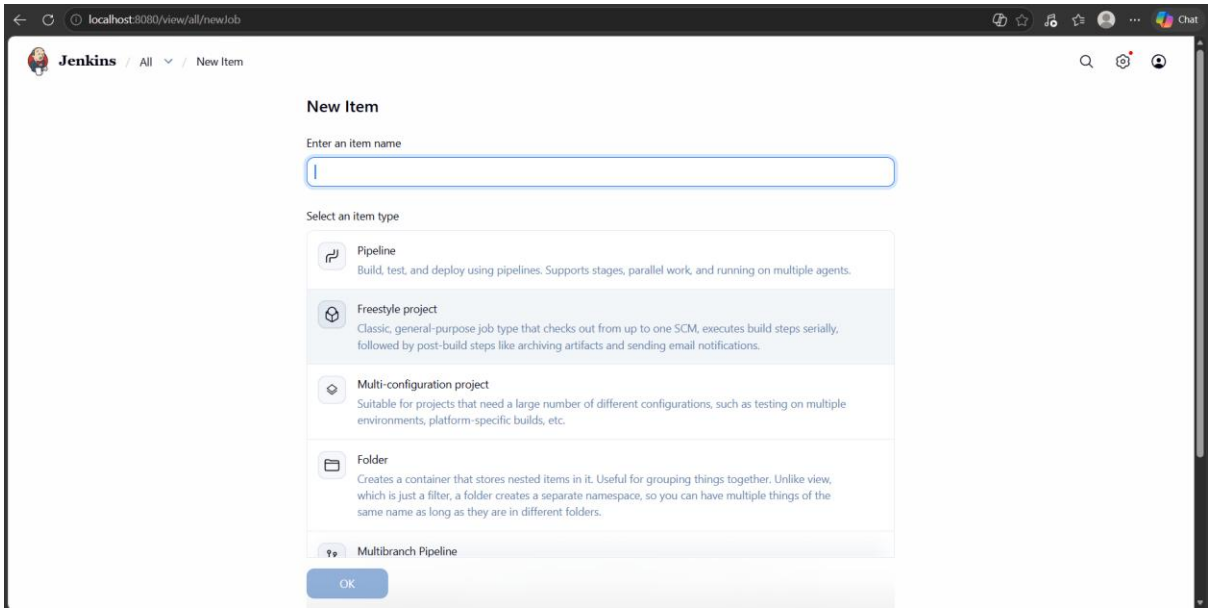

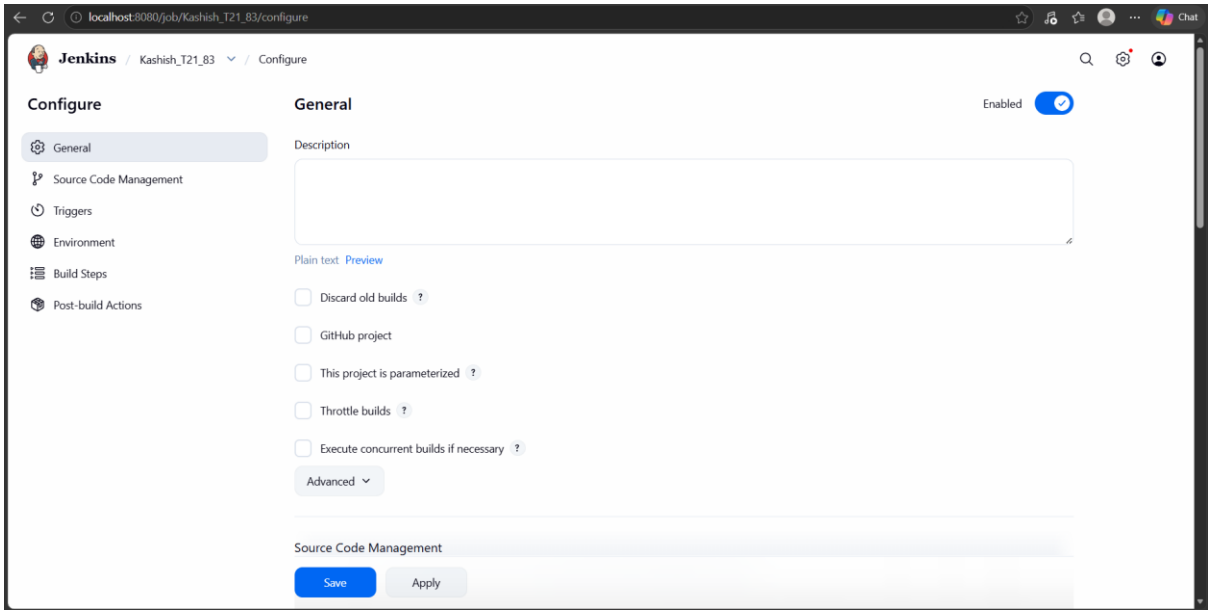

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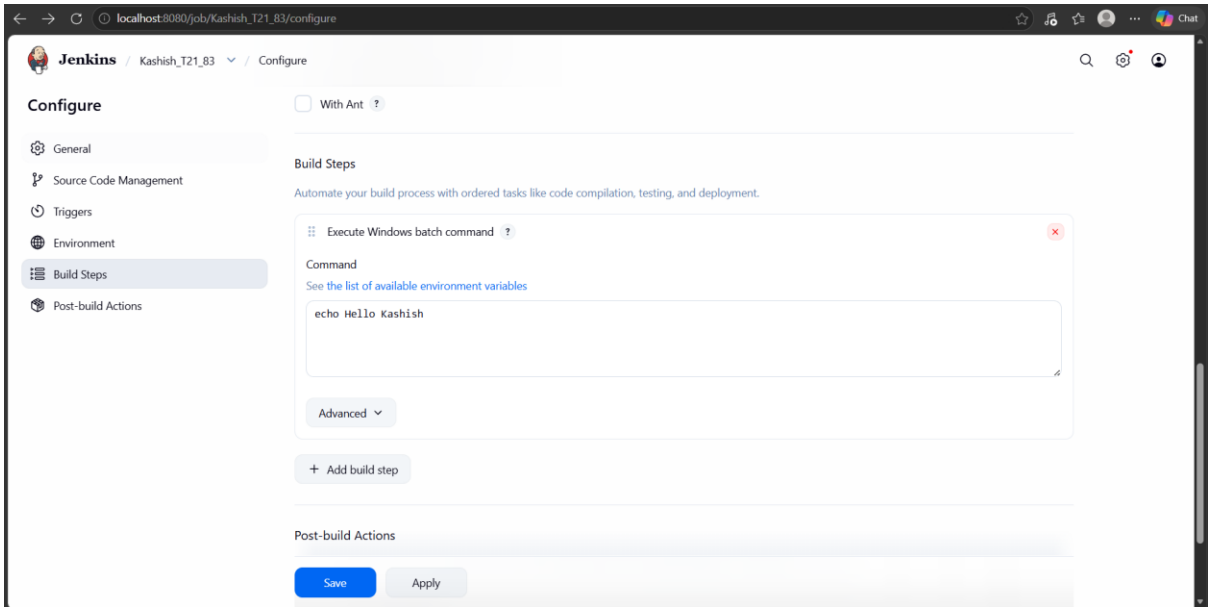

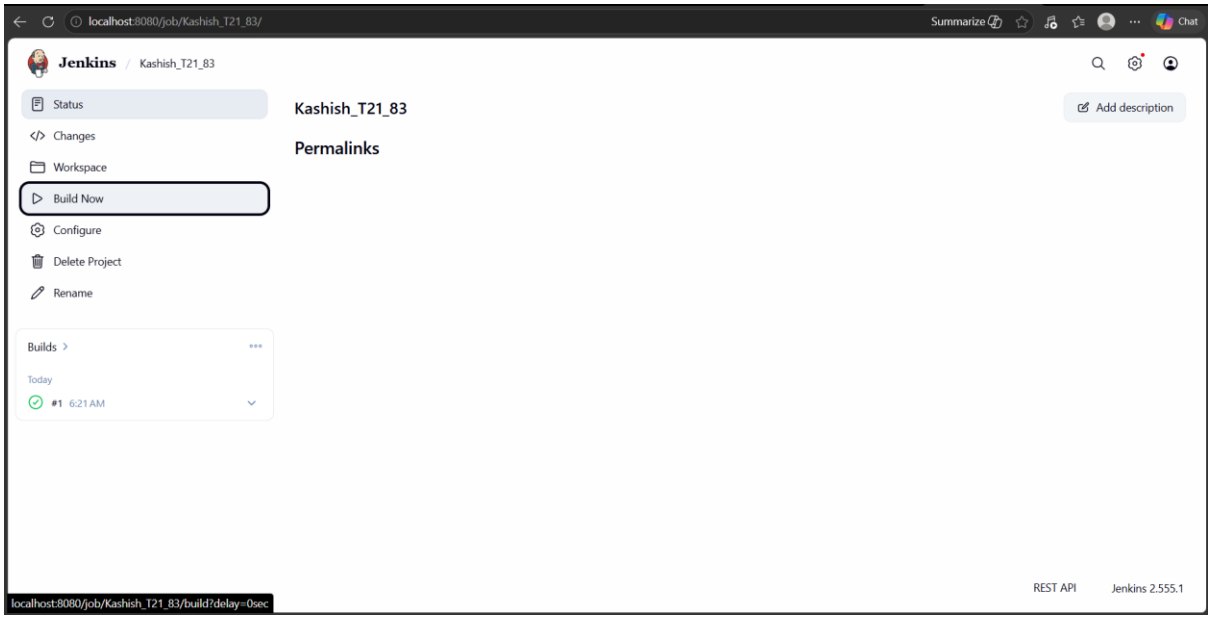

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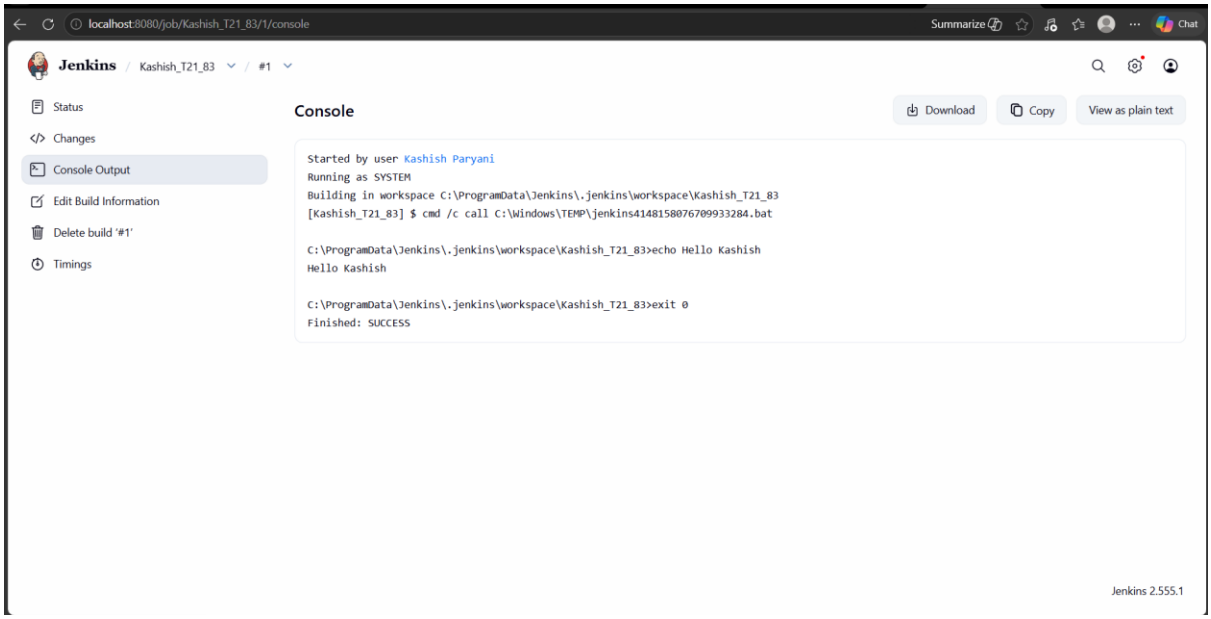

Conclusion: Successfully implemented the pipeline of jobs using Maven / Gradle / Ant in Jenkins, create a pipeline script to Test and deploy an application over the tomcat server